TreasuryAudit360 – Demo Video Narration Script

For the 6 min 12 s demo video (TreasuryAudit360_Demo.mp4) – record your voice-over against these timings

Workflow automation for audit of the Treasury department of a bank in India

Regulatory coverage: RBI | SEBI | IFSCA (GIFT City) | FEMA

Auditing framework: ICAI Standards on Auditing and Standards on Internal Audit

Platform: n8n Cloud | Claude (Anthropic) | Google Sheets | Gmail | Lovable

Capstone project submitted by: **CA Ravishankar Balakrishnan Aiyar**

Membership No.: **040524**

Batch: **AIL2B105**

Date: **30 October 2026**

Contents

1. Where the app is hosted and published 3

2. AI tools used – at every instance 3

3. Programming languages and technologies – at every instance 3

4. Slide-by-slide script (timings match the MP4) 4

5. Recording tips 5

1. Where the app is hosted and published

What	Link / location
Live web app (public, hosted on Lovable)	https://treasuryaudit360.lovable.app
Web app source and editor (Lovable)	https://lovable.dev/projects/81783906-7834-4b2a-b9f3-a475c9424449
Audit Engine webhook (n8n Cloud, published)	https://nonchallant.app.n8n.cloud/webhook/treasury-audit/run (POST)
Findings / Data / Response API (n8n Cloud, published)	https://nonchallant.app.n8n.cloud/webhook/treasury-audit/findings · /data · /response
n8n workflows (sign in to view)	https://nonchallant.app.n8n.cloud/workflow/UAWiMJKX6g0aDfu6 (Engine) · https://nonchallant.app.n8n.cloud/workflow/T1Z2qgsCgpxW9gob (API) · https://nonchallant.app.n8n.cloud/workflow/lIUnOcDT7XaVmba2 (Setup)
Google Sheet (data, findings, audit trail)	https://docs.google.com/spreadsheets/d/1HogxieIBSdBVLSRC7ktPiUNNHaB2ruNFZ6teT0Rkuek/edit

2. AI tools used – at every instance

AI tool	Exactly where it was used
Claude Opus 5.5 (Claude Code, Claude desktop app)	Designed the solution; wrote the 3 n8n workflows as TypeScript SDK code and published them through the n8n MCP connector; wrote the 23 JavaScript audit tests; created the fictitious bank data; wrote the Lovable build prompt and published the app through the Lovable MCP connector; tested every webhook; produced the guide (Markdown, Word, PDF), the screenshots, slides and this video
Claude Sonnet 5.5 (Anthropic)	Runs inside n8n (Anthropic node, n8n AI credits) on every audit run; drafts the Audit Committee report from the computed JSON facts
Lovable AI agent	Generated the 7-page React web app from the prompt and hosts it on lovable.app
Rule engine (not AI)	Every finding is decided by deterministic JavaScript rules; AI never decides a finding

3. Programming languages and technologies – at every instance

Where	Language / technology	Purpose
n8n workflow definitions	TypeScript (n8n Workflow SDK)	Nodes, connections, groups and sticky notes of all 3 workflows
Setup workflow – Seed nodes	JavaScript (n8n Code node)	Fictitious deals, investments, counterparties, dealers, positions, restricted list, parameters
Audit Engine – Run Audit Tests	JavaScript (n8n Code node)	23 tests, risk rating, compliance matrix, ICAI SA list, SA 530 sample, data fingerprint
Audit Engine – Assemble Report	JavaScript + HTML/CSS	Cleans Claude output, builds Annexures A and B, email subject, API response
Normalize Request / Audit Trail / Response nodes	n8n expressions ({{ }} – JavaScript)	Field mapping and defaults
Draft Audit Report	English system prompt → Claude Sonnet 5.5	Prompt engineering: role, SIA 370 / SA 265 structure, "use only the facts"
All API calls	JSON over HTTPS (REST webhooks)	Web app ↔ n8n; JSON brief to Claude
Data store	Google Sheets API (OAuth2)	7 data tabs, Findings register, Audit_Runs trail
Email	Gmail API (OAuth2), HTML	Report emailed to the auditor
Web app	React (TSX), TypeScript, Tailwind CSS, shadcn/ui, Recharts	Dashboard, Run Audit, Report, Findings, Compliance, Data Explorer, Architecture
Guide document	Markdown → Python (python-docx), Word export to PDF	Treasury_Audit_Automation_Guide (.md / .docx / .pdf)
Demo video	PowerShell + headless Microsoft Edge (Chrome DevTools Protocol) for screenshots; Python (Pillow) for slides; PowerPoint (COM) for MP4	This video and the editable deck

4. Slide-by-slide script (timings match the MP4)

Time	Slide	On screen	What to say
0:00	1	Title	Introduce yourself and the project: an end-to-end automated audit of a bank's treasury, covering RBI, SEBI, IFSCA and FEMA, documented per ICAI Standards on Auditing.
0:07	2	Where the app is hosted and published	Show the hosting layers: the public web app on Lovable, the n8n Cloud webhooks, the Google Sheet, the GitHub Pages version and the Android app.
0:19	3	What is CAAT?	Explain CAAT: computer-assisted audit techniques test every record with software. Here the 23 rules are the CAATs, giving 100 percent coverage under SA 330 and SA 530.
0:29	4	Audit scope and objectives	Explain the coverage across four regulators and how the engagement is mapped to the ICAI Standards on Auditing.
0:40	5	AI tools used – and where	Walk through each AI tool. Stress the control: findings come from deterministic rules; Claude Sonnet only drafts the narrative from computed facts, and the auditor reviews it.
0:54	6	Programming languages and technologies – at every layer	Explain each language at each layer: TypeScript for workflow definitions, JavaScript for the audit tests, JSON for the APIs, HTML for the report, React and TypeScript for the web app, and Python and PowerShell for the documents and this video.
1:08	7	End-to-end architecture	Narrate the full loop from data extraction to testing, AI drafting, documentation, email, the web app and the management response loop.
1:20	8	About / Architecture page in the app	Show the architecture page in the app and the four endpoints.
1:27	9	Data 1 Deal blotter – 26 deals	This is the population tested: 26 treasury deals across money market, G-Sec, forex, derivatives, equity and the IBU.
1:35	10	Data 2 Investment register – HTM / AFS / FVTPL	The investment register supports the HTM, valuation, non-SLR and depository reconciliation tests.
1:42	11	Data 3 Daily risk positions	Daily positions feed the NOOP, call money and capital market exposure tests.
1:49	12	Data 4 Parameters – regulation-aware without code changes	Emphasise that every regulatory threshold is a parameter, so the tool can be updated to the latest circulars without changing code.
1:58	13	n8n Workflow 0 – Setup demo data (run once)	Workflow zero is a one-time setup that creates the workbook and seeds the data.
2:05	14	n8n Workflow 1 – Audit Engine (published)	Walk node by node: trigger, normalise input, read seven tabs, run the 23 JavaScript tests, Claude drafts the report, assemble annexures, save findings and the SA 230 audit trail, email the report and respond to the web app.
2:19	15	Inside the rule engine – JavaScript in the n8n Code node	Show sample rules. Each rule is a few lines of JavaScript, reads its threshold from the Parameters tab and tags the exception with the regulation and ICAI standard.
2:31	16	Claude Sonnet 5.5 – system prompt (prompt engineering)	Explain the prompt design: role, standards, required sections, HTML output, and the instruction to use only the supplied facts.
2:43	17	n8n Workflow 2 – Findings & Data API (published)	Workflow two is the API the web app uses: read findings, read the data population and write management responses back.
2:53	18	Step 1 Dashboard	Open the dashboard: rating, findings, amount flagged and charts, plus the history of audit runs.
3:02	19	Step 2 Run a new audit	Go to Run Audit, enter your name and click Run Audit.
3:10	20	Step 3 Audit in progress (about 60–120 seconds)	While the stages run, explain what n8n is doing at each stage.
3:19	21	Step 4 Audit result	Read out the result: rating, tests, findings, population and amount flagged.
3:28	22	Step 5 Regulatory compliance matrix	Explain the compliance matrix by regulator and the ICAI standards applied.
3:37	23	Step 6 23 tests on 100% of the population (CAAT)	Scroll the tests performed, showing population and exceptions for each test.
3:45	24	Step 7 SA 530 walkthrough sample	Explain that full-population testing replaces statistical sampling, but a walkthrough sample is still selected for vouching.
3:54	25	Step 8 Audit Committee report drafted by Claude	Open the Audit Report page and read the executive summary.

Time	Slide	On screen	What to say
4:03	26	Step 9 Compliance matrix and key observations	Scroll to the observations grouped by regulator.
4:13	27	Step 10 SA 265 deficiencies and follow-up	Show the SA 265 section, the follow-up table and the Print to PDF button.
4:23	28	Step 11 Findings Register	Open the Findings Register and filter by regulator, for example IFSCA or SEBI.
4:31	29	Step 12 Finding TD006 and the management response loop	Open finding TD006, where the dealer confirmed their own deal, read the details, then enter a management response and submit. Show the Findings tab in Google Sheets updated.
4:42	30	Step 13 Compliance Matrix page	Show the compliance matrix page with a card per regulator.
4:49	31	Step 14 Tests grouped by regulator	Scroll through the tests grouped by regulator.
4:57	32	Step 15 Regulatory framework covered	Read out the regulatory framework covered and the note to verify thresholds against current circulars.
5:06	33	Step 16 GitHub-ready app (single file, installable)	Show the GitHub version: one HTML file, no build step, installable as an app, with the n8n workflows and sample data in the same repository.
5:15	34	Step 17 The Android app	Show the Android app on a phone: the same dashboard and findings register, full screen in the navy theme. Mention the APK for direct install and the AAB for the Play Store.
5:27	35	How the Android app was built (no Android Studio)	Explain the build: resources compiled with aapt2, Java compiled and converted to Android bytecode, aligned and signed into an APK, and bundled into an AAB for Google Play.
5:38	36	Audit test catalogue (23 tests)	Summarise the 23 tests and their regulator and ICAI standard mapping.
5:49	37	Benefits, controls and limitations	Close with benefits, the controls over AI, the limitations and future scope.
6:02	38	Thank you	Thank the panel and give the live link.

5. Recording tips

- Play the MP4 full-screen and record your voice with the Windows **Xbox Game Bar** (Win + Alt + R) or **Clipchamp** (built into Windows: import the MP4, add a voice-over track).
- Alternatively open **TreasuryAudit360_Demo.pptx**, use **Slide Show → Record** to narrate slide by slide; the "Say:" note for each slide is in the speaker notes. Then **File → Export → Create a Video**.
- For a live segment, open https://treasuryaudit360.lovable.app and do one real run (Step 2–4) – it takes about 1–2 minutes; cut the wait in editing.
- The n8n canvases in the video are diagrams. If you want the real canvas, sign in to n8n and record the Audit Engine workflow for 15–20 seconds.